



POLITECNICO
MILANO 1863

**MULTIMODAL INTERACTIVE
TECHNOLOGIES AND
APPLICATIONS**

Project Concept

Botanical Garden

Group Name	Gardeners
Kaifei Xu	11115439
Keyan Cheng	11154362
Shiqi Huang	11111452
Yufei Liu	11118413

May 25, 2026

1 Overview

Botanical Garden is a browser-based botanical garden that combines **immersive panoramic exploration** with a **conversational, voice-driven assistant**. Users move through 360° garden scenes, look at or select 3D plant models, and ask natural-language questions. The goal is to turn a static botanical exhibit into a fluid multimodal experience where visual context, plant selection, speech input, adaptive information panels and TTS feedback work together.

2 Characterizing Features

- **Immersive 3D environment.** The prototype renders linked 360° panoramic garden scenes with navigation portals and 3D plant models. This gives users a spatial context for exploration.
- **Multimodal input.** The system combines visual context, plant selection, typed input, speech-to-text via the Web Speech API, and click/touch fallback controls. Users can therefore interact by looking, selecting, speaking, typing, or clicking overlay options depending on the situation.
- **Multimodal output.** Responses are presented through a coordinated combination of 3D plant highlights, adaptive 2D information panels (botanical, medicinal, comparison and fallback), and text-to-speech narration.
- **LLM-assisted dialogue with local fallback.** A lightweight stateless backend provides JSON endpoints for intent parsing, information generation, disambiguation prompts and comparison speech. The prototype first attempts LLM-assisted generation when enabled, but keeps local deterministic parsing and structured plant data as fallback so that the core interaction does not break when the LLM is unavailable.
- **Referential ambiguity handling.** When the user asks an ambiguous question such as “*what is this*” while multiple plants are visible or otherwise plausible in the current scene, the system enters a referential ambiguity flow. Candidate plants are labelled A / B / C and can be resolved by clicking, typing, or speaking a letter or plant name, with retry guidance and direct-click fallback.
- **Contextual comparison.** The system remembers recently visited plants and can compare the current selection with a named plant, a previously visited plant, or a plant found in the global database. In the current prototype, comparison focuses on drought tolerance while the panel also shows supporting attributes where available.

3 Target Users and Why It Is Multimodal

The system targets **botanical garden visitors, botany students and users who are unable to visit in person** exploring a botanical collection on desktop. Instead of reading long static labels, users can ask focused questions about the plant in front of them.

Botanical Garden is multimodal **by design**: users interact through multiple input modalities, including visual context, click/raycast selection, typed text and speech input, while the system responds through multiple output modalities, including 3D spatial highlights, visual information cards and spoken TTS feedback.